Tutorial on Agentic Graph Learning: Foundations and Frontiers

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Abstract

Graph learning has been widely adopted in real-world applications to effectively capture complex structural dependencies and semantic information from graph-structured data. However, traditional methods, such as graph neural network (GNN)-based methods, face two fundamental limitations: they typically apply one-size-fits-all message-passing mechanisms to all nodes despite substantial structural and semantic heterogeneity, and they often require graph-specific training, resulting in limited transferability and generalization across different graphs and tasks. In response, Agentic Graph Learning (AGL) has recently emerged as a promising new paradigm for graph learning. By leveraging the reasoning and planning capabilities of large language models, AGL formulates graph learning as a sequential decision-making process, where agents autonomously plan their interactions with graph environments and progressively acquire task-relevant structural and semantic evidence through graph-native tools. In this tutorial, we aim to provide a comprehensive introduction to AGL, spanning its foundations and recent frontiers. We will cover: (1) the fundamental concepts, motivations, and key building blocks of agentic graph learning; (2) a systematic taxonomy of existing AGL methods, including single-agent graph reasoning, multi-agent collaboration, and their associated training strategies and applications; and (3) the key limitations, open challenges, and future research directions of AGL. Through this tutorial, we aim to provide a clear foundation for AGL and inspire future research and applications.

CCS Concepts

• **Theory of computation → Multi-agent learning**.

Keywords

Agentic Graph Learning, Graph Learning, Large Language Models, AI Agents, Multi-Agent Systems

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1 Outline

Graph learning [16] has been widely adopted in real-world applications to effectively capture complex structural dependencies and semantic information from graph-structured data. For example, in social networks, users and their interactions can be represented as graphs, allowing learning methods to exploit both connectivity patterns and user information for downstream tasks [12]. In recommender systems, user–item interactions are naturally modeled as bipartite graphs, where graph learning can capture collaborative signals beyond direct interactions to improve personalized recommendation [8, 14]. Similar graph structures also arise in academic citation networks [2] and molecular data [17], where graph learning can effectively capture structural information encoded in the relationships among entities.

Traditional graph learning is largely dominated by graph neural network (GNN)-based methods [9, 10, 15]. The core idea of GNNs is to iteratively aggregate information from neighboring nodes and propagate it through the graph to learn expressive node or graph representations. Early representative models include Graph Convolutional Networks (GCNs) [5], GraphSAGE [4], and Graph Attention Networks (GATs) [13], which differ mainly in how neighborhood information is aggregated and weighted. More recently, graph learning has increasingly adopted Transformer-based architectures to capture longer-range dependencies and improve representation capacity [7]. Despite their success, these methods still face two important limitations. First, they typically rely on a largely fixed, one-size-fits-all information propagation mechanism, which may not adequately account for substantial structural and semantic heterogeneity across nodes. Second, they usually require graph-specific training or adaptation, leading to limited transferability and generalization across different graphs and tasks.

To address these limitations, recent graph learning research has begun to explore a new direction centered on Agentic Graph Learning (AGL) [1]. Rather than treating graph learning as a fixed message-passing process, AGL formulates it as a sequential decision-making process in which large language model-based agents can autonomously interact with graph environments, determine what information to acquire, and progressively collect task-relevant evidence through graph-native tools. Existing AGL methods can be broadly organized into two major categories. Single-agent AGL focuses on how an individual agent explores graph structure, gathers relevant evidence, and performs graph reasoning for downstream tasks [3, 11]. Multi-agent AGL extends this paradigm by introducing multiple specialized agents that communicate, coordinate, and collaborate over graph-structured information [6, 18]. These two directions are further connected with different training and optimization strategies, as well as a growing range of applications across graph learning tasks.

This tutorial will focus specifically on the foundations, methodologies, and emerging frontiers of AGL. We will first introduce the motivation, problem formulation, and key building blocks of agentic graph learning, with background to make the tutorial accessible to participants familiar with machine learning or graph learning but new to agentic AI. We will then provide an in-depth taxonomy of existing methods, covering single-agent graph reasoning, multi-agent collaboration, and their associated training strategies and applications, with representative methods used to illustrate the design choices and technical differences within each category. Finally, we will discuss the current limitations of AGL, and highlight open problems and promising future research directions. The tutorial is designed to provide both a systematic foundation for newcomers and a technical overview of recent advances for researchers interested in developing the next generation of agentic graph learning methods.

2 Goals and Objectives

Overall, this tutorial aims to benefit the participating audience from the following three aspects:

- We aim to provide participants with a comprehensive and up-to-date picture of Agentic Graph Learning, enabling them to understand the current development, representative methodologies, and emerging trends in this rapidly evolving area.
- We will present a systematic taxonomy of existing AGL methods, helping participants develop a structured understanding of the field. Representative single-agent and multi-agent approaches will be discussed in detail, together with their associated training strategies and applications.
- We will discuss the key limitations, open challenges, and promising future directions of AGL, helping participants identify important research opportunities and potential avenues for further advancing this emerging field.

3 Schedule

The tutorial is planned for 1.5 hours and consists of five sections. In what follows, we provide a detailed outline of the tutorial.

Section 1. Welcome and Introduction (10 mins) **Presenter: Prof. Jianxin Li** 1.1 Overview of Graph Learning and Its Applications 1.2 Motivation for Agentic Graph Learning
Section 2. Definition and Taxonomy of Agentic Graph Learning (15 mins) **Presenter: Prof. Jianxin Li** 2.1 Problem Formulation of Agentic Graph Learning 2.2 Key Building Blocks of Agentic Graph Learning 2.3 Taxonomy of Existing AGL Methods
Section 3. Agentic Graph Learning Methods (40 mins) **Presenter: Dr. Liang Qu** 3.1 Single-Agent Graph Learning 3.2 Multi-Agent Graph Learning
Section 4. Limitations, Challenges, and Future Directions (15 mins) **Presenter: Dr. Liang Qu**
Section 5. Open Discussion (10 mins) **Presenters: Prof. Jianxin Li and Dr. Liang Qu**

4 Intended Audience

This tutorial is designed for participants from both academia and industry who are interested in emerging directions in graph learning and in understanding how agentic techniques are reshaping the field. Prior exposure to basic machine learning concepts will be helpful, but no previous experience with agentic graph learning is required, as the tutorial will introduce the necessary background before moving to more advanced topics. By the end of the tutorial, participants are expected to have a clear understanding of how AGL differs from traditional graph learning, how current methods are organized, and where the most important research opportunities lie.

5 Tutorial team

Prof. Jianxin Li and Dr. Liang Qu have extensive research experience in graph learning and agentic AI, with a strong publication record in leading international conferences and journals.

- **Prof. Jianxin Li** is a Professor with extensive research experience in information systems, data science, artificial intelligence, and graph-related learning problems. He has published around 200 peer-reviewed research papers in leading international conferences and journals, including SIGMOD, PVLDB, ICDE, WWW, KDD, IEEE TKDE, Knowledge-Based Systems, and Information Systems. A number of his publications have been recognized as Highly Cited Papers or Hot Papers. He has secured more than ten externally funded research projects with a total value of approximately AUD 1.7 million, including Australian Research Council Discovery and Linkage Projects. He has more than 16 years of teaching experience across several Australian universities, covering undergraduate and postgraduate courses in data science, artificial intelligence, and business intelligence. He currently serves as Associate Editor for several international journals, including Information Systems, Knowledge-Based Systems, IEEE Signal Processing Letters, and World Wide Web. He has also served in major conference leadership roles, including Program Committee Chair or Steering Committee Member for ADMA 2019, APWeb-WAIM 2023, and numerous international workshops.
- **Dr. Liang Qu i**s a Lecturer in the School of Information and Communication Technology at Griffith University. He received his PhD from the School of Electrical Engineering and Computer Science at The University of Queensland in 2024 and subsequently worked as a Postdoctoral Research Fellow at Edith Cowan University in 2025. His research focuses on graph learning, agentic AI, and recommender systems. He has published more than 40 research papers in leading journals and conferences in artificial intelligence, data mining, and information retrieval, including IEEE TKDE, ACM TOIS, WWW, KDD, and SIGIR. His recent research explores agent-based learning and collaboration over graph-structured data, which is closely aligned with the central topic of this tutorial. He currently serves as an Associate Editor of Array and regularly reviews papers for major international journals and conferences.

6 Tutorial materials

Upon acceptance of the tutorial, the presentation slides and other supporting materials will be made available to participants through our tutorial website (https://github.com/Leo-Q-316/ADMA2026-Agentic-Graph-Learning-Tutorial) before the conference.

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